VNC Remote Control

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        Reconnection Symptom
        Solution

This tutorial applies to systems shipped with JetPack 7.2.1 / Ubuntu 24.04. It shares the local Jetson desktop through `x11vnc`, and Windows connects with VNC Viewer.

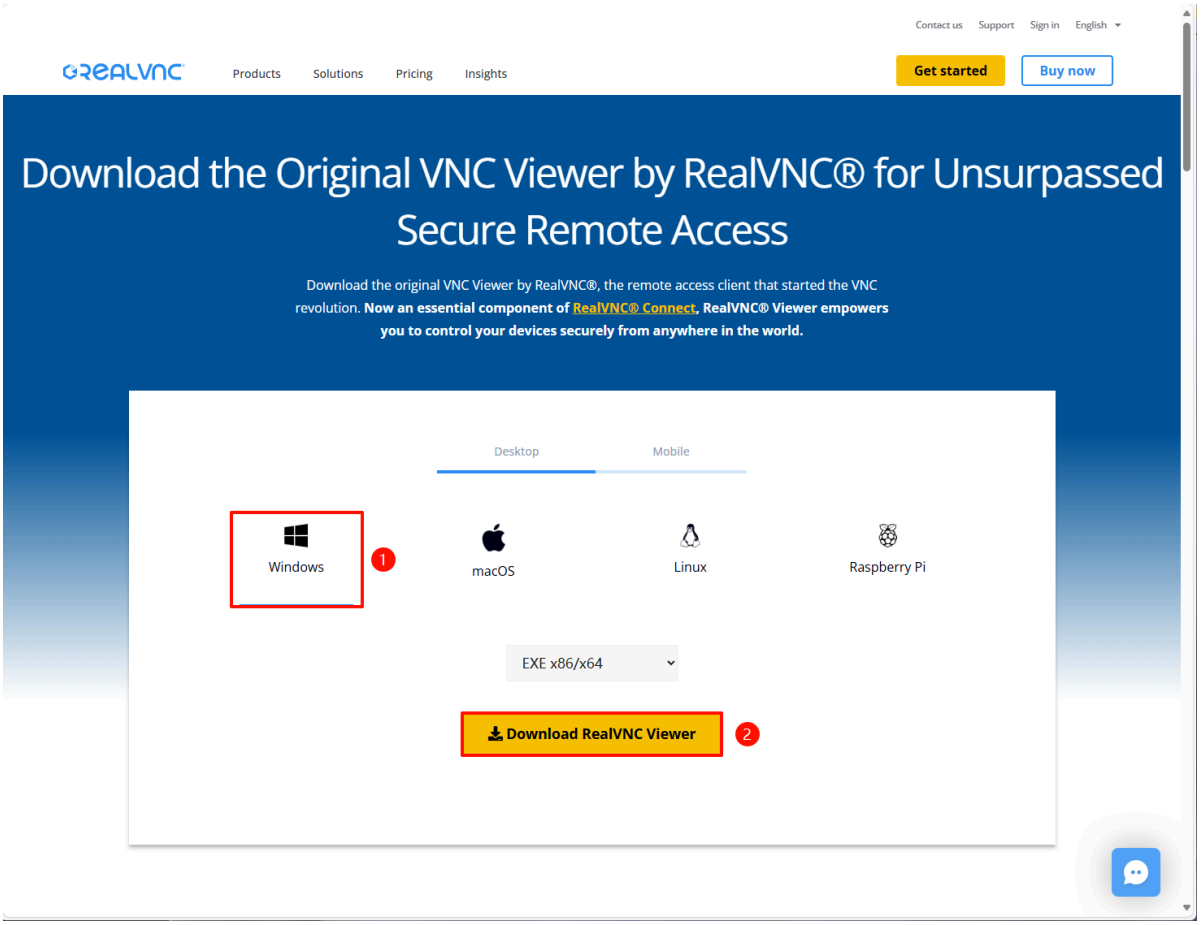
> The "Settings → System → Remote Desktop" of Ubuntu 24.04 uses RDP and cannot serve as the server of VNC Viewer. This tutorial no longer uses the "Sharing → Remote Desktop → Legacy VNC" settings of Ubuntu 22.04. When RDP remote login is required, refer to RDP Remote Control.

The Windows computer needs VNC Viewer installed in advance, and must be on the same local area network as the Jetson.

1. VNC Viewer

1.1. Download VNC

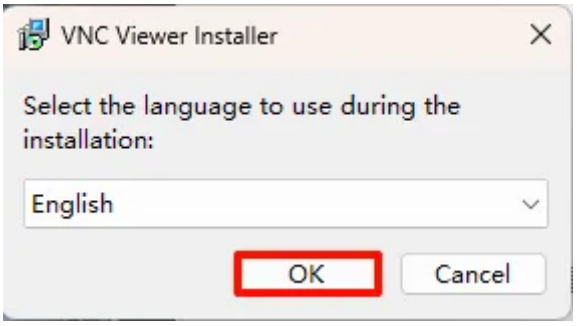
Official download website: https://www.realvnc.com/en/connect/download/viewer/

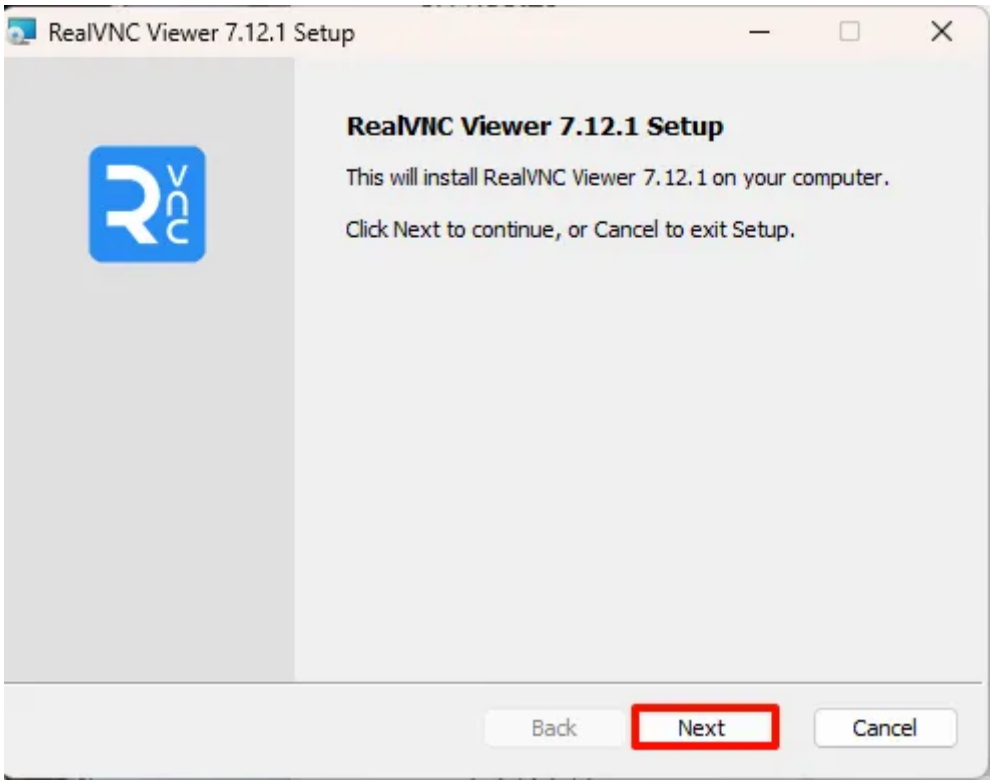


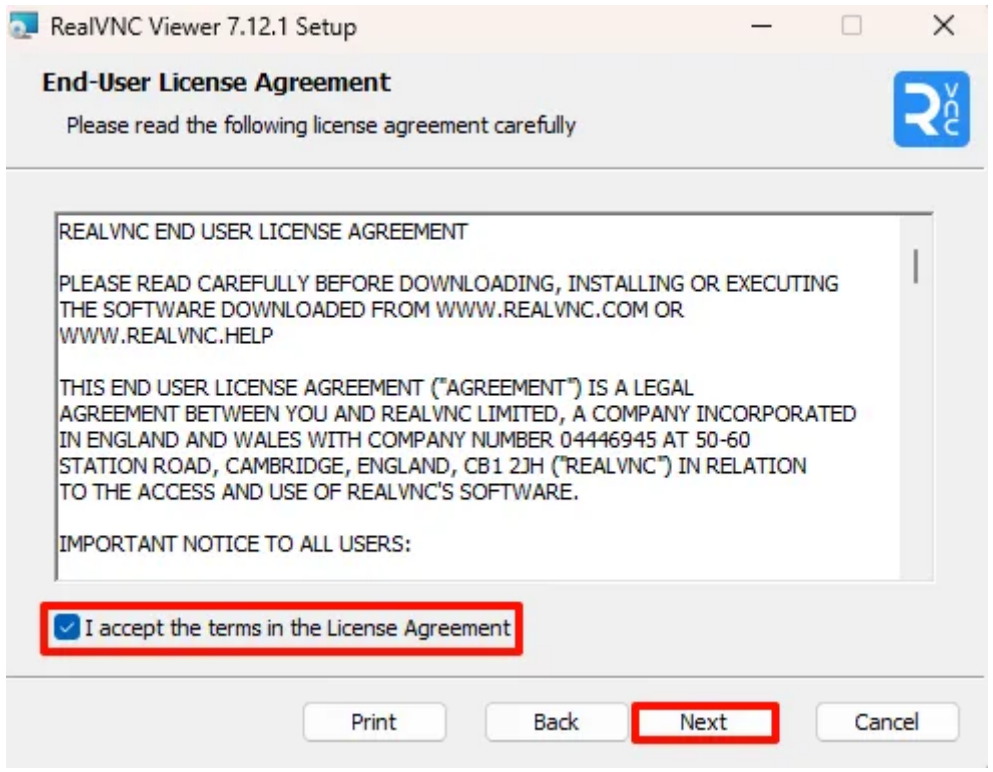
1.2. Install VNC

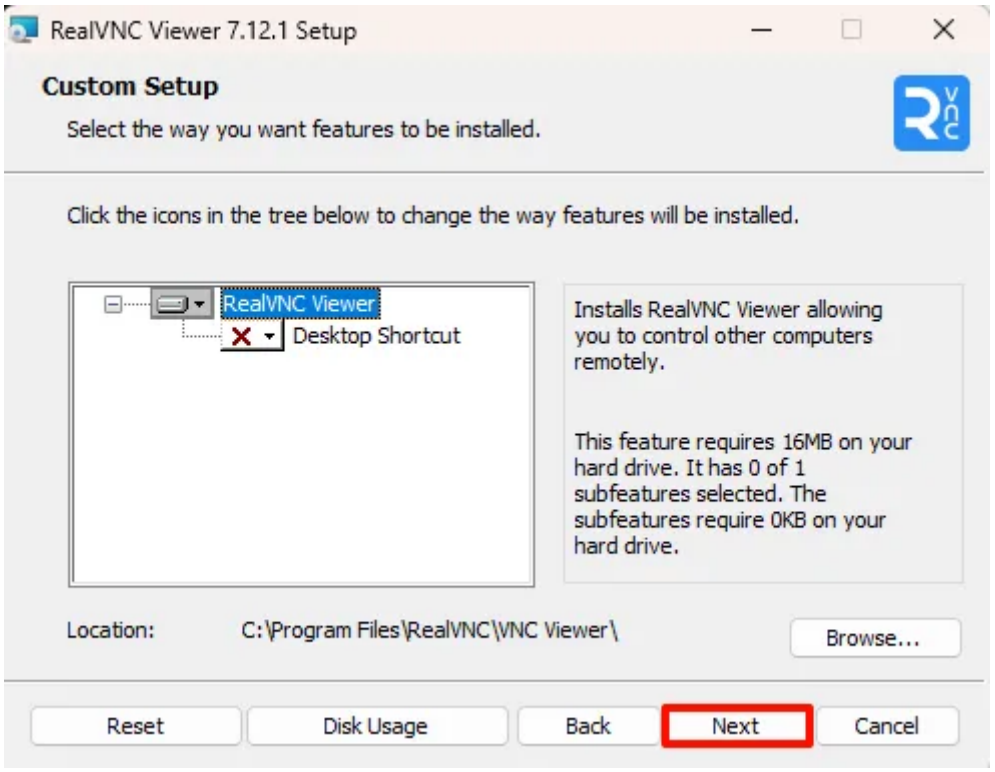
Run VNC-Viewer-xxx.exe as administrator:

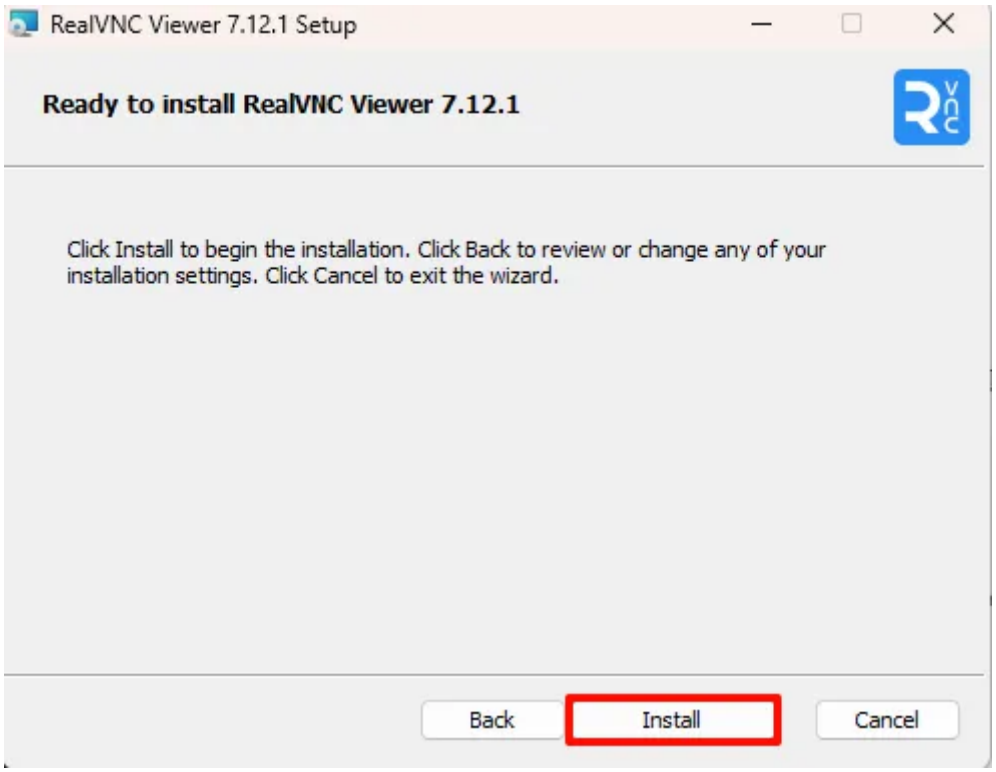


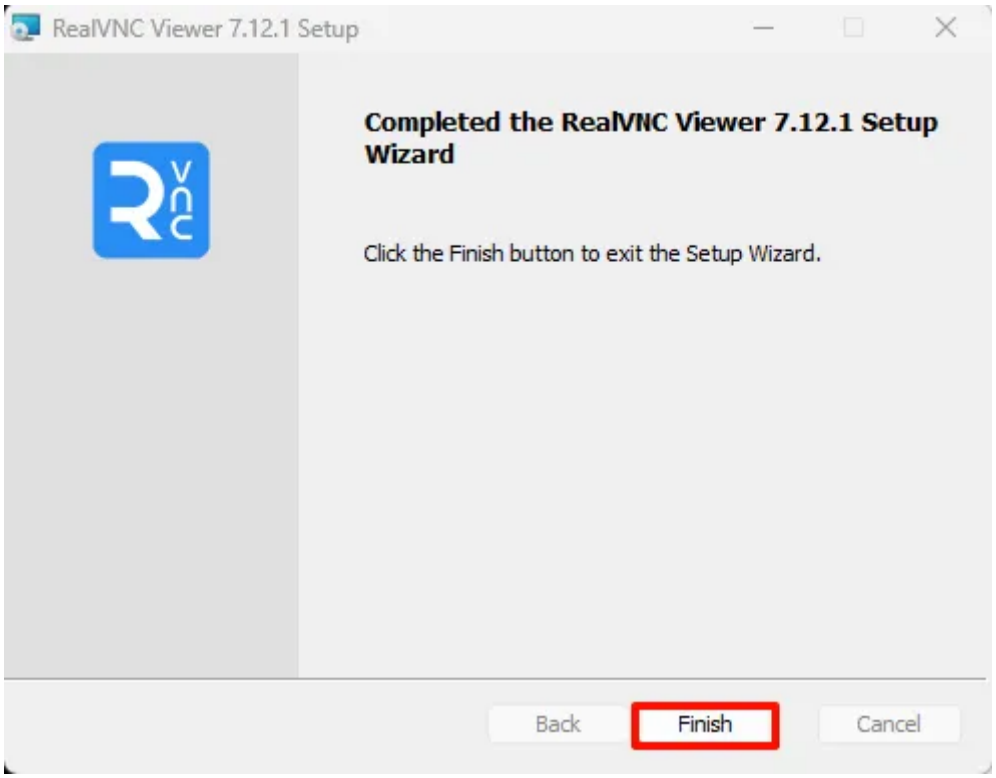



RealVNC Viewer 7.12.1 Setup
RealVNC Viewer 7.12.1 Setup
This will install RealVNC Viewer 7.12.1 on your computer.
Click Next to continue, or Cancel to exit Setup.
Back
Next
Cancel

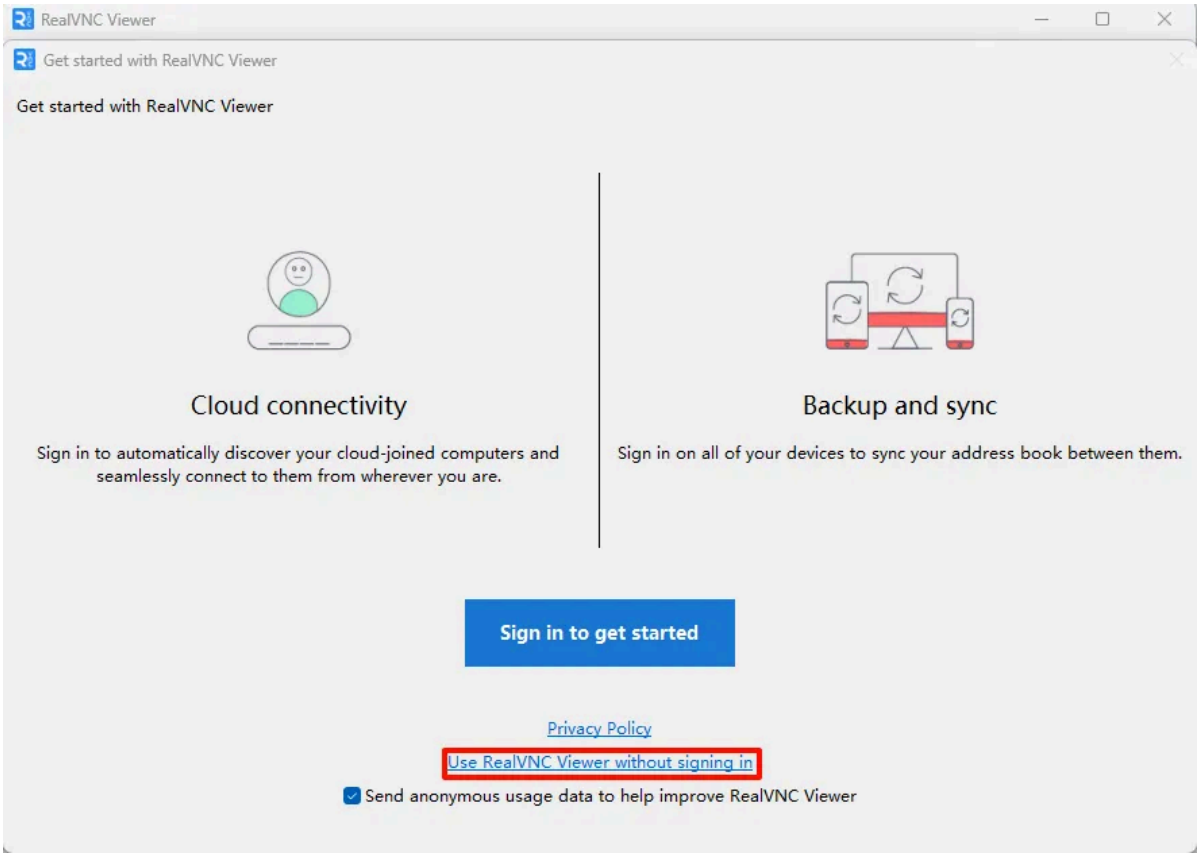

RealVNC Viewer 7.12.1 Setup
End-User License Agreement
Please read the following license agreement carefully
REALVNC END USER LICENSE AGREEMENT
PLEASE READ CAREFULLY BEFORE DOWNLOADING, INSTALLING OR EXECUTING THE SOFTWARE DOWNLOADED FROM WWW.REALVNC.COM OR WWW.REALVNC.HELP
THIS END USER LICENSE AGREEMENT ("AGREEMENT") IS A LEGAL AGREEMENT BETWEEN YOU AND REALVNC LIMITED, A COMPANY INCORPORATED IN ENGLAND AND WALES WITH COMPANY NUMBER 04446945 AT 50-60 STATION ROAD, CAMBRIDGE, ENGLAND, CB1 2JH ("REALVNC") IN RELATION TO THE ACCESS AND USE OF REALVNC'S SOFTWARE.
IMPORTANT NOTICE TO ALL USERS:
I accept the terms in the License Agreement
Print
Back
Next
Cancel

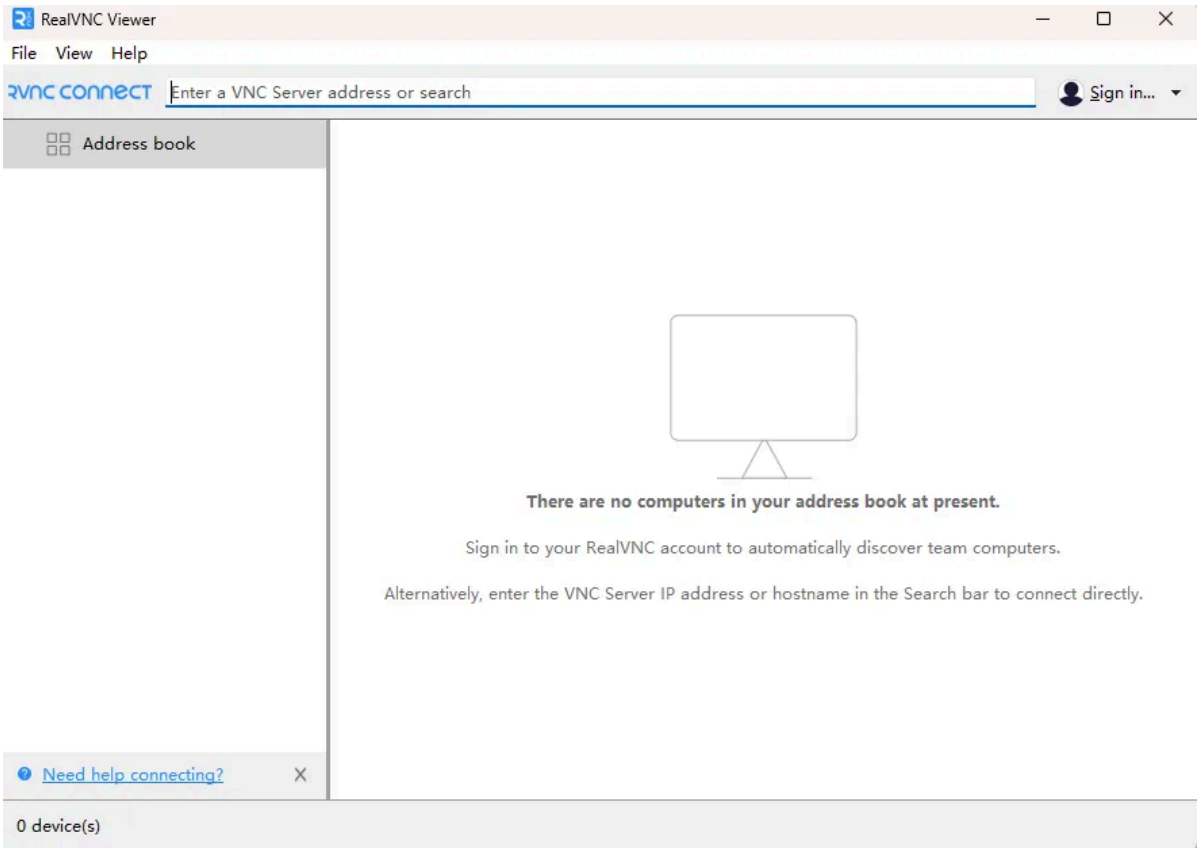

RealVNC Viewer 7.12.1 Setup
Custom Setup
Select the way you want features to be installed.
Click the icons in the tree below to change the way features will be installed.
RealVNC Viewer
Desktop Shortcut
Installs RealVNC Viewer allowing you to control other computers remotely.
This feature requires 16MB on your hard drive. It has 0 of 1 subfeatures selected. The subfeatures require 0KB on your hard drive.
Location:
C:\Program Files\RealVNC\VNC Viewer\
Browse...
Reset
Disk Usage
Back
Next
Cancel


RealVNC Viewer 7.12.1 Setup
Ready to install RealVNC Viewer 7.12.1
Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.
Back
Install
Cancel


RealVNC Viewer 7.12.1 Setup
Completed the RealVNC Viewer 7.12.1 Setup Wizard
Click the Finish button to exit the Setup Wizard.
Back
Finish
Cancel

1.3. Using VNC


RealVNC Viewer
Get started with RealVNC Viewer
Get started with RealVNC Viewer
Cloud connectivity
Sign in to automatically discover your cloud-joined computers and seamlessly connect to them from wherever you are.
Backup and sync
Sign in on all of your devices to sync your address book between them.
Sign in to get started
Privacy Policy
Use RealVNC Viewer without signing in
Send anonymous usage data to help improve RealVNC Viewer



2. System Settings (Jetson)

2.1. Switch to VNC Mode

The shipped system has already configured `x11vnc` and the remote desktop mode switching script. Execute the following in the Jetson terminal or over SSH:

```
sudo switch-remote-desktop vnc
```

This command restores the local automatic login and the VNC shared desktop mode, and restarts `gdm3`. The switch interrupts the current graphical session; save your work in desktop programs before the operation, and confirm that backup connections such as SSH are available.

View the current mode:

```
sudo switch-remote-desktop status
```

> `switch-remote-desktop` is a script provided by the shipped system, not a built-in Ubuntu command. If you write the official NVIDIA clean system yourself, this script and the VNC service may be missing, so `x11vnc` needs to be installed and configured separately, and the switching steps of the shipped system cannot be copied directly.

2.2. Confirm the Desktop and VNC Service

The current VNC solution shares the local X11 desktop that is already logged in on the Jetson, usually the `:0` session on the monitor. It does not create an independent remote desktop; when the local desktop has not been logged in yet, the graphical session has exited, or Wayland is in use, a direct connection cannot be made in this way.

Check the service and the listening port in the Jetson:

```
systemctl status x11vnc.service --no-pager
ss -ltn | grep ':5900'
```

The service should show `active (running)` and listen on TCP `5900`. If the service is not running, check the cause:

```
journalctl -u x11vnc.service -b -n 50 --no-pager
```

Provided that the Jetson has entered the local desktop, you can try restarting the service and then check the status again:

```
sudo systemctl restart x11vnc.service
```

Do not configure VNC by enabling the "Media Sharing" of Ubuntu; media sharing has nothing to do with VNC remote control. There is also no need to install the `allow-locked-remote-desktop` extension or clear the keyring password of "Passwords and Keys" as the old tutorials describe.

2.3. VNC Password and Connection at Boot

VNC Viewer uses the **VNC access password** configured in `x11vnc`, which is not necessarily the same as the Jetson system login password. Use the VNC password provided by the shipped system; do not use the credentials on the Ubuntu RDP page as a substitute.

If you need to troubleshoot the password configuration or change the password, first check the service startup parameters:

```
systemctl cat x11vnc.service
```

If `-rfbauth` appears in the startup parameters, the path following it is the password file used by the service; when changing the password, operate on this actual file. Do not assume that the password of the existing service has been changed merely because you ran `x11vnc -storepasswd` to generate another file.

Automatic connection after boot requires all of the following at the same time: the `x11vnc` service is enabled, the local X11 desktop is logged in, and TCP `5900` is reachable. Check whether the service is set to start at boot:

```
systemctl is-enabled x11vnc.service
```

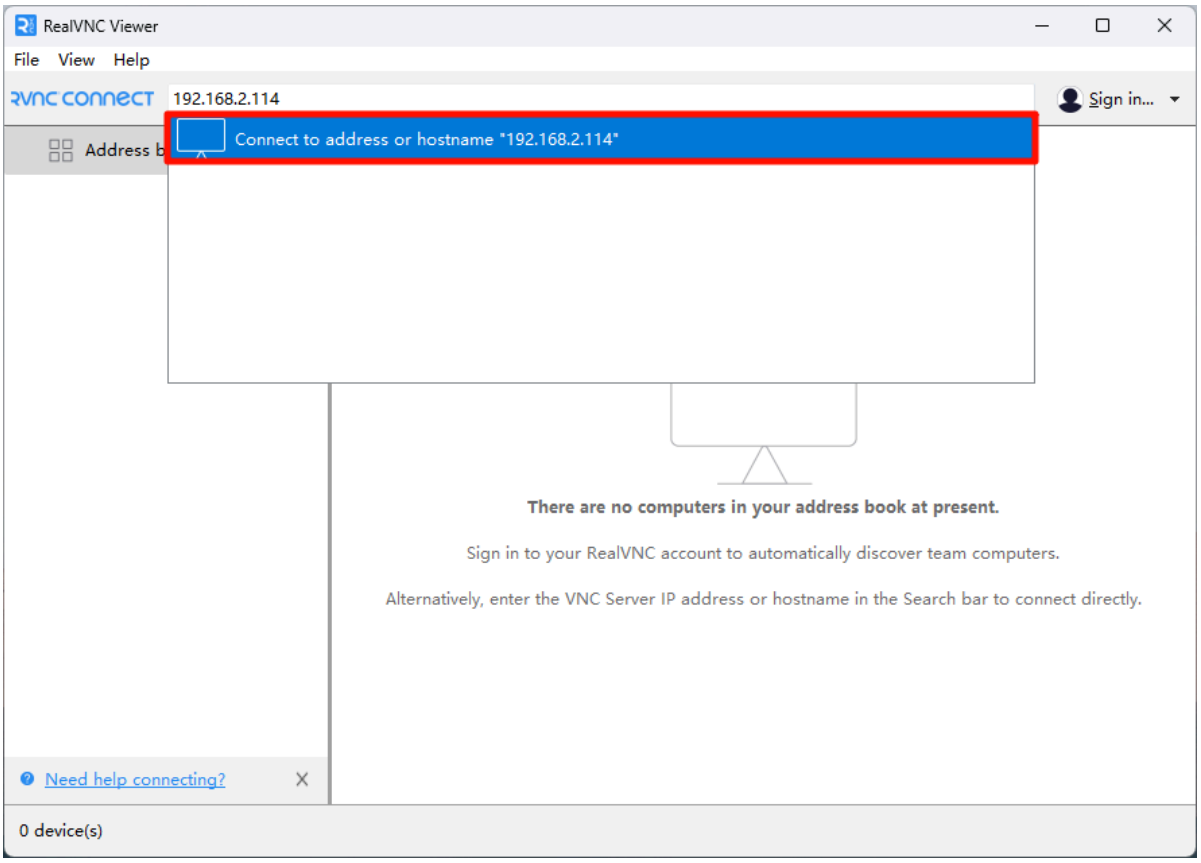
If it shows `disabled` and the service has been verified as available in the previous section, you can enable it:

```
sudo systemctl enable x11vnc.service
```

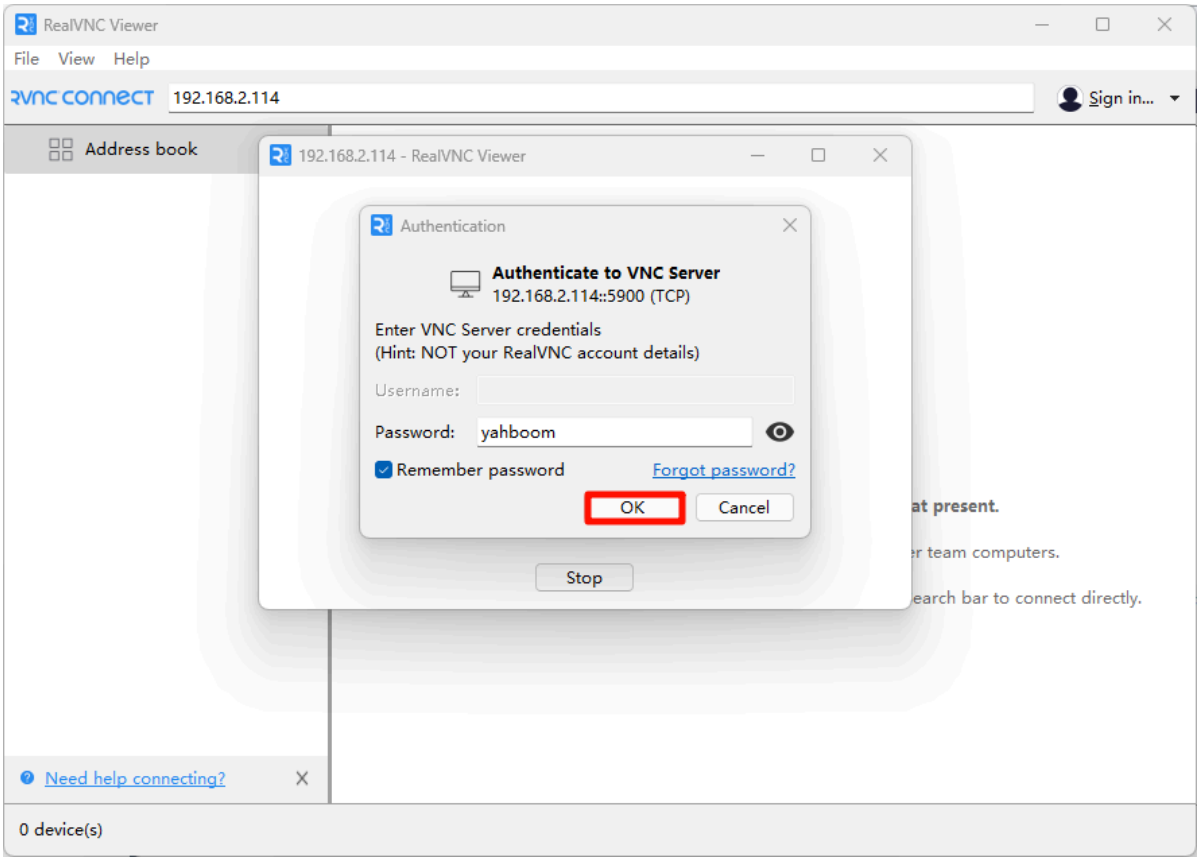
Enabling the service alone cannot replace the local desktop login. The VNC mode of the shipped system depends on local automatic login; after switching to RDP mode, automatic login is turned off, and you need to switch back to VNC mode to restore the original way of use. The connection behavior after the screen is locked should be confirmed on the current image; do not apply the effect of the old GNOME extension to `x11vnc`.

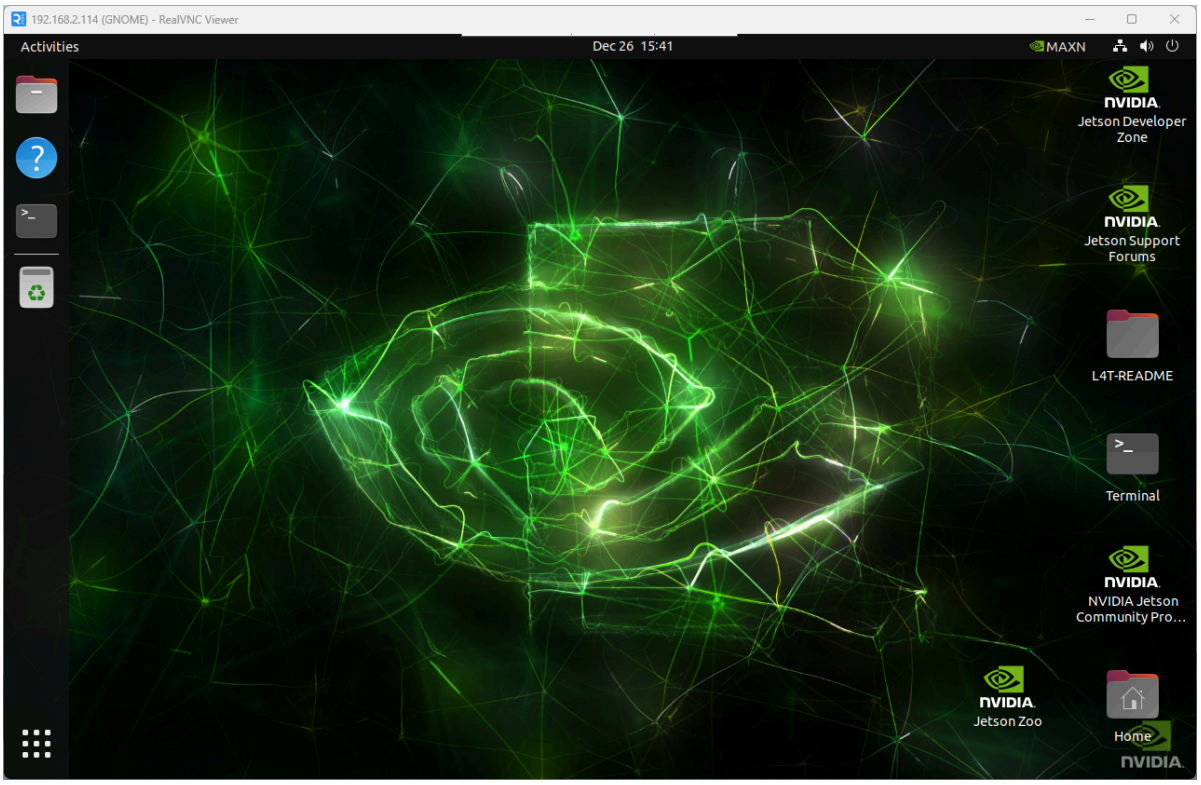
3. VNC Remote Control

Run `hostname -I` in the Jetson terminal to view the mainboard IP. Enter this IP in VNC Viewer; TCP `5900` is used by default, and you can also enter `IP:5900` (replace `IP` with the actual address).



Enter the VNC access password as prompted (see Section 2.3); do not use the system login password directly. The screenshots below only illustrate the client connection process; for the address and credentials in the old screenshots, refer to the current device.





Common Problems

Cannot Find the VNC Switch of Remote Desktop

The built-in remote desktop of Ubuntu 24.04 uses RDP. "Settings → System → Remote Desktop" cannot enable the `x11vnc` service of this tutorial; on the shipped system, use the mode switching command in Section 2.

Message `switch-remote-desktop: command not found`

Confirm whether the current system is the shipped system or the official clean system installed by yourself. This command is a shipped script; if it is missing, it cannot be made up by installing GNOME extensions. First check whether the `x11vnc` service has been installed and configured; if `systemctl status x11vnc.service` reports that the service does not exist, the VNC server needs to be configured separately.

Cannot Connect or Black Screen After Connection

First confirm that the IP of the computer and the Jetson are reachable, then check the service and the `5900` listening in Section 2.2. Confirm that VNC mode has been switched back, the Jetson local desktop is logged in, and the network or firewall is not blocking `5900`. When no monitor is connected, you also need to confirm that the system has actually established a shareable X11 desktop; do not assume that VNC is available merely because SSH works.

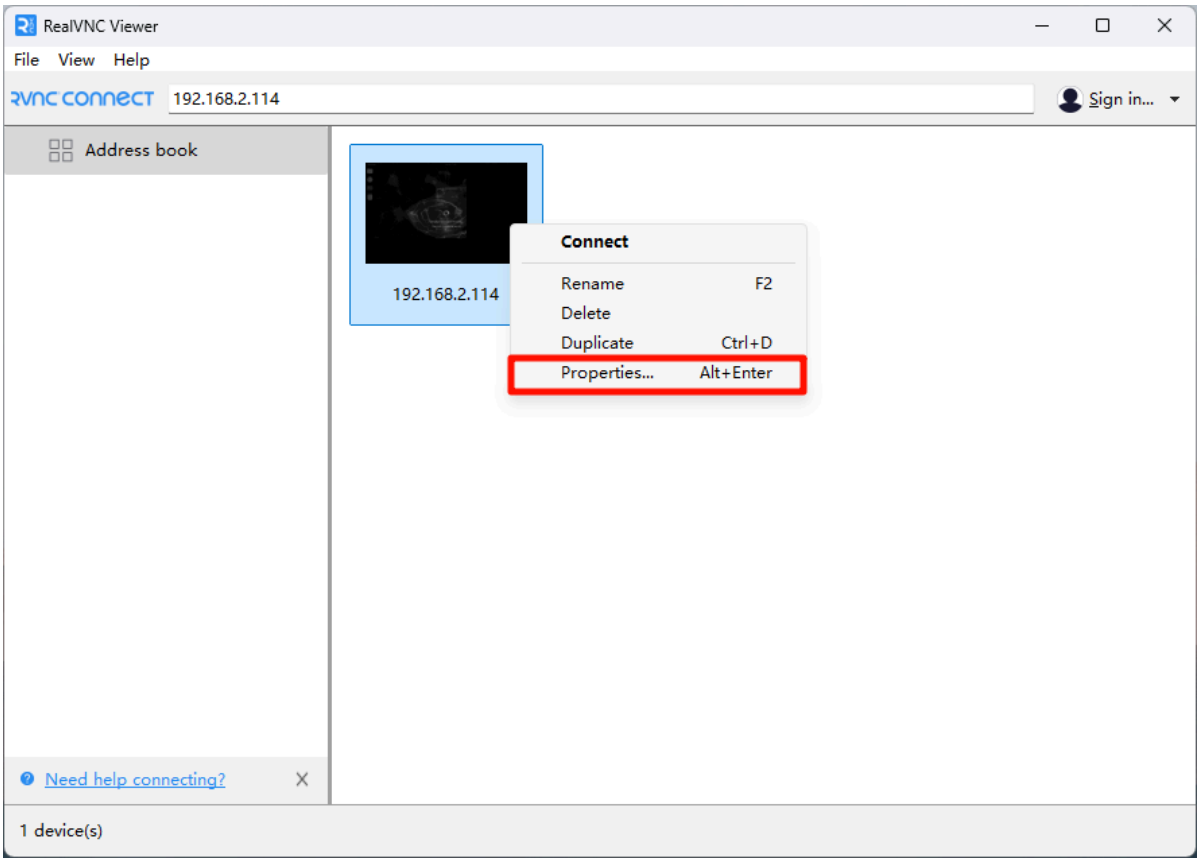
VNC Remote Display Reconnection

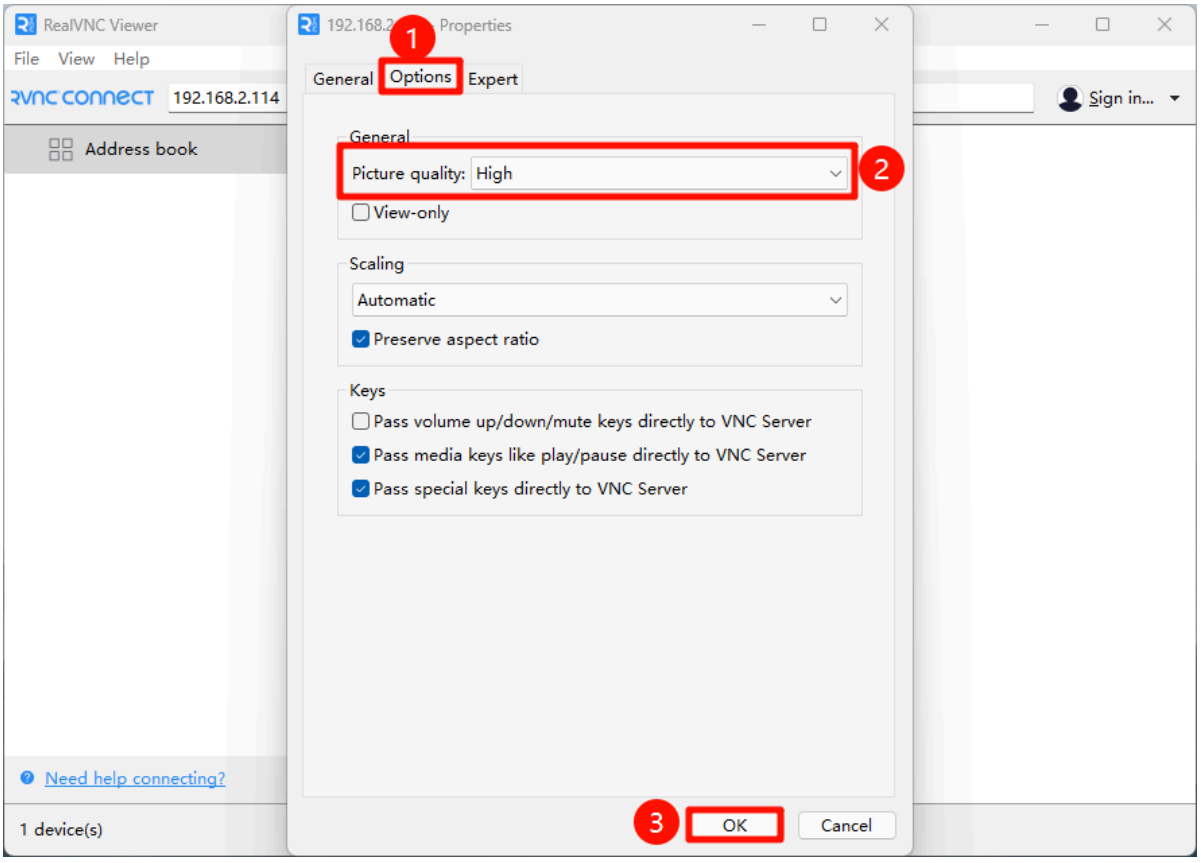
Reconnection Symptom

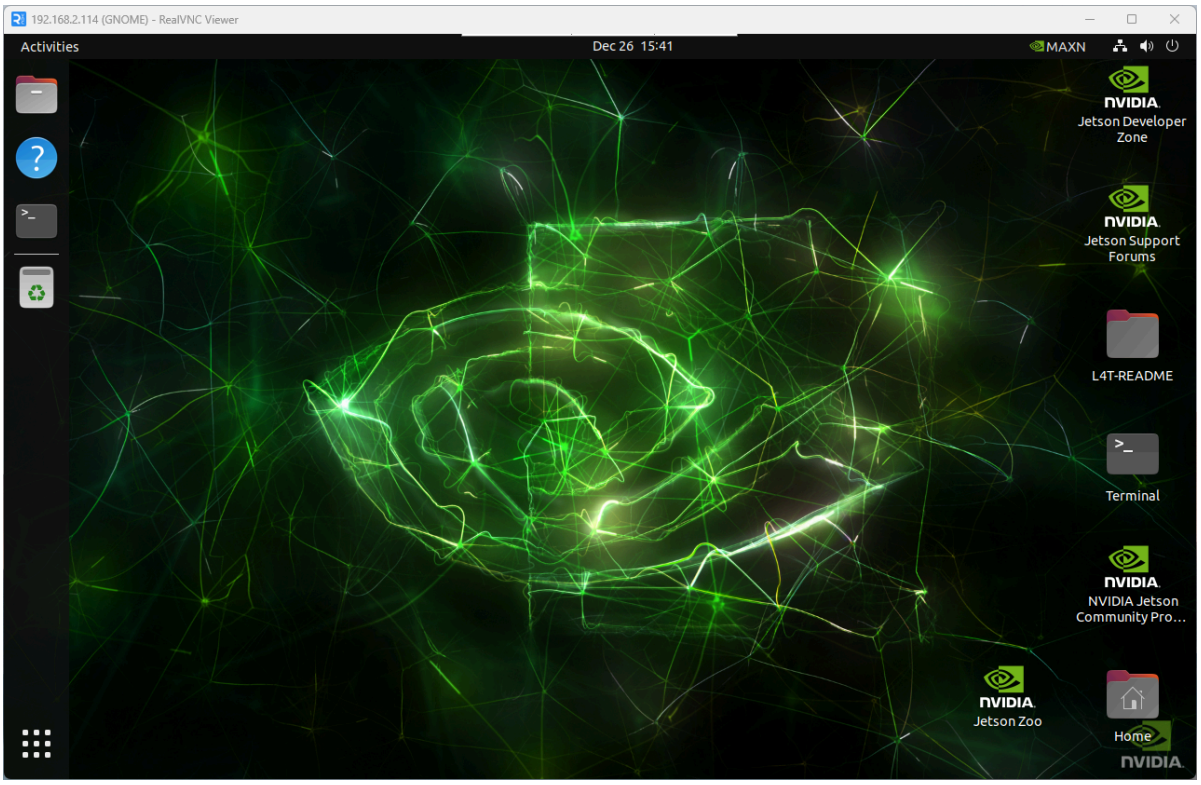


Solution

Modify the options of the corresponding remote device → specify the remote picture quality


RealVNC Viewer
File View Help
VNC CONNECT
192.168.2.114
Sign in...
Address book
192.168.2.114
Connect
Rename F2
Delete
Duplicate Ctrl+D
Properties... Alt+Enter
Need help connecting?
1 device(s)


RealVNC Viewer
File View Help
VNC CONNECT
192.168.2.114
Sign in...
Address book
Need help connecting?
1 device(s)
Properties
1
General
Options
Expert
General
Picture quality: High
2
View-only
Scaling
Automatic
Preserve aspect ratio
Keys
Pass volume up/down/mute keys directly to VNC Server
Pass media keys like play/pause directly to VNC Server
Pass special keys directly to VNC Server
3
OK
Cancel


192.168.2.114 (GNOME) - RealVNC Viewer
Activities
Dec 26 15:41
MAXN
NVIDIA Jetson Developer Zone
NVIDIA Jetson Support Forums
L4T-README
Terminal
NVIDIA NVIDIA Jetson Community Pro...
NVIDIA Jetson Zoo
Home
NVIDIA